Extra Exercises

Chapter 14

(1) Find the gradient vector ∇f of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at the point $(1, 2, 2)$.

(2) Suppose that over a certain region of space the electrical potential V is given by

$$V(x, y, z) = xy \sin z.$$

(a) Find the gradient vector of V at the point $P\left(1, 2, \frac{\pi}{2}\right)$.

(b) Find the rate of change of V at $P\left(1, 2, \frac{\pi}{2}\right)$ in the direction of the vector $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

(c) In which direction does V change most rapidly at P?

(d) What is the maximum rate of change at P?

(3) Given $f(x, y) = 4y^3 + x^2 - 12y^2 - 36y + 2$.

(a) Find all the critical points of f.

(b) Use the second derivative test to classify the nature of each critical point.

(c) Find the local extreme of f, if it exists.

(4) Suppose $(0, 2)$ is a critical point of a function g with continuous second-order partial derivatives. In each case, what can you say about g?

(a) $g_{xx}(0, 2) = -1$, $g_{xy}(0, 2) = 6$, $g_{yy}(0, 2) = 1$.

(b) $g_{xx}(0, 2) = -1$, $g_{xy}(0, 2) = 2$, $g_{yy}(0, 2) = -8$.

(5) Use Lagrange multipliers to find the minimum value of $f(x, y, z) = 2x + 2y + 2z + 2$ subject to the constraint $9x^2 + 4y^2 - 2z = 0$.

(6) Find the minimum and maximum distances from the sphere $x^2 + y^2 + z^2 = 4$ to the point $(3, 1, -1)$.

Note: Questions (5) and (6) do not belong to COM211.

Chapter 15

(1) Evaluate the following iterated integrals.

(a) $\displaystyle\int_0^1 \int_x^1 \cos\left(y^2\right)\ dydx.$

(b) $\displaystyle\iint_R \frac{\sin x}{x}\ dA$, where R is the triangle bounded by the lines $y = 0$, $y = x$, $x = 1$.

(c) $\displaystyle\int_0^2 \int_0^3 \int_0^4 dxdydz.$

(d) $\displaystyle\int_0^\pi \int_0^1 \int_0^{\sqrt{1-y^2}} y \sin x\ dzdydx.$

(2) Find the area enclosed by the lemniscate $r^2 = 4\cos 2\theta$.

(3) Find the volume of the solid that lies under the surface $z = 4 - x^2 - y$ and above the rectangle $R = [0, 1] \times [0, 2]$.

(4) Use cylindrical coordinates to evaluate $\displaystyle\iiint_E (x - y)\ dV$, where E is the solid that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 16$.

(5) Use spherical coordinates to evaluate $\displaystyle\iiint_E (x^2 + y^2)\ dV$, where E lies between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.

(6) Use the transformation $u = x + y$, $v = x - y$ to evaluate the integral

$$\iint_R (x + y)e^{(x+y)(x-y)}\ dA,$$

where R is the rectangle enclosed by the lines $x - y = 0$, $x - y = 2$, $x + y = 0$, and $x + y = 3$.

Note: Question (6) does not belong to COM211.

Chapter 16

(1) Evaluate the line integral, where C is the given curve.

(a) $\displaystyle\int_C (x/y)\, ds, \quad C : x = t^3, y = t^4, 1 \le t \le 2.$

(b) $\displaystyle\int_C xe^y\, ds, \quad C$ is the line segment from $(2, 0)$ to $(5, 4)$.

(2) Let $\mathbf{F}(x, y, z) = \sin x\mathbf{i} + \cos y\mathbf{j} + xy\mathbf{k}$ be a vector field. Evaluate the line integral $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is given by the vector function $\mathbf{r}(t) = t^3\mathbf{i} - t^2\mathbf{j} + t\mathbf{k}$, $0 \le t \le 1$.

(3) Evaluate the line integral $\displaystyle\oint_C 4x^2y dx + 2y dy$, where C is the boundary of the triangle with vertices $(0, 0)$, $(1, 2)$, $(0, 2)$, oriented positively.

(4) Let $\mathbf{F}(x, y, z) = x^5\mathbf{i} + y^5\mathbf{j} + z^5\mathbf{k}$ be a vector field.

(a) Show that $\mathbf{F}$ is a conservative vector field.

(b) Find a potential function f for $\mathbf{F}$.

(5) Let $\mathbf{F}(x, y, z) = \langle e^x \sin y, e^y \sin z, e^z \sin x\rangle$ be a vector field.

(a) Find the curl of $\mathbf{F}$.

(b) Find the divergence of $\mathbf{F}$.

(c) Determine whether or not $\mathbf{F}$ is conservative. If it is conservative, find a function f such that $\mathbf{F} = \nabla f$.

(6) The parametric representation for the torus obtained by rotating about the z-axis the circle in the xz-plane with center $(b, 0, 0)$ and radius $a < b$ is $\mathbf{r}(\alpha, \theta) = x(\alpha, \theta)\mathbf{i} + y(\alpha, \theta)\mathbf{j} + z(\alpha, \theta)\mathbf{k}$ where $x = b\cos\theta + a\cos\alpha\cos\theta$, $y = b\sin\theta + a\cos\alpha\sin\theta$, $z = a\sin\alpha$. Find the surface area of the torus.

